



HYDERABAD INSTITUTE OF TECHNOLOGY AND MANAGEMENT

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II -II CSM SECTION-A

PROBLEM BASED LEARNING

TEAM NO:12

ABSTRACT

Sentiment analysis is an important technique used to understand customer opinions from product reviews. It helps organizations know whether customers feel positive, negative, or neutral about a product. In this project, a machine learning–based system is developed to automatically classify customer sentiment using textual product reviews.

The project uses both traditional machine learning algorithms and modern deep learning models for sentiment classification. Traditional algorithms such as Support Vector Machine (SVM) and Decision Tree are implemented using text feature extraction methods like Bag-of-Words and TF-IDF. These methods convert text into numerical form so that machine learning models can process it. Traditional models are simple, fast, and easy to implement, but they struggle to understand the context and meaning of words in longer sentences.

To overcome these limitations, deep learning models such as Recurrent Neural Networks (RNN) and Transformer-based models are used. RNN models analyze text in a sequence, which helps in understanding word order and sentence structure. Transformer models use attention mechanisms to focus on important words in a review, allowing them to capture deeper contextual meaning and relationships between words more effectively.

All models are evaluated and compared using performance metrics such as accuracy, precision, and recall. The results show that deep learning models provide better sentiment classification performance than traditional machine learning algorithms. However, traditional models require fewer computational resources and work well for smaller datasets. This project highlights the strengths and limitations of different approaches and helps in selecting the appropriate model for customer sentiment analysis.



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